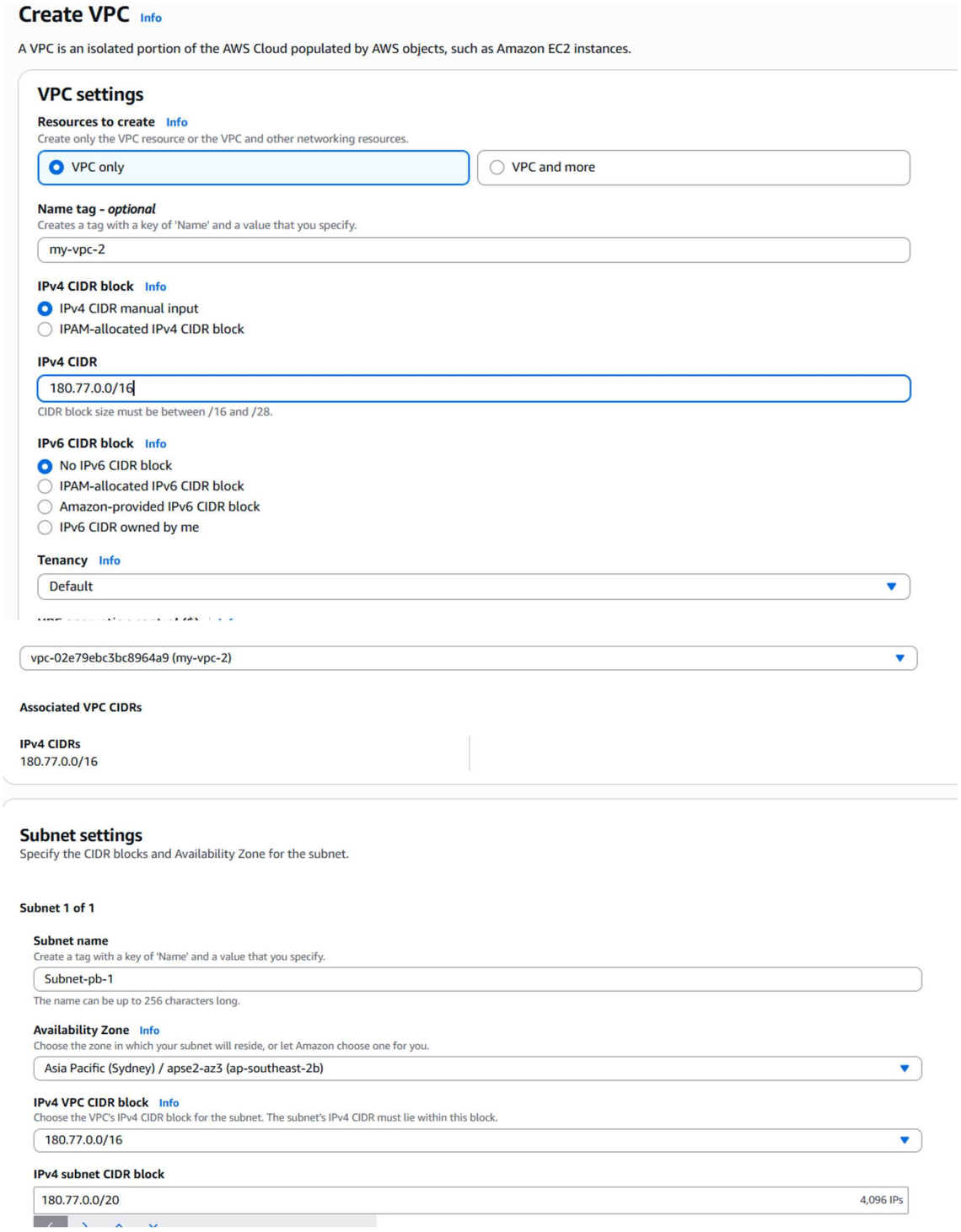
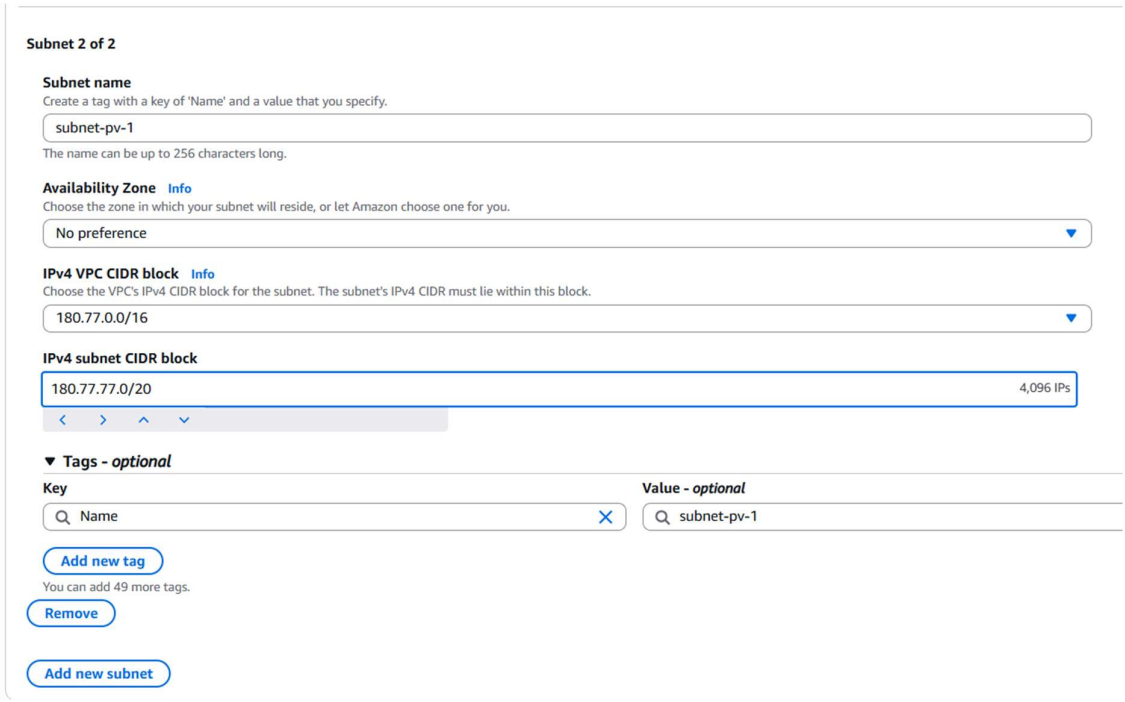
VPC Task-2

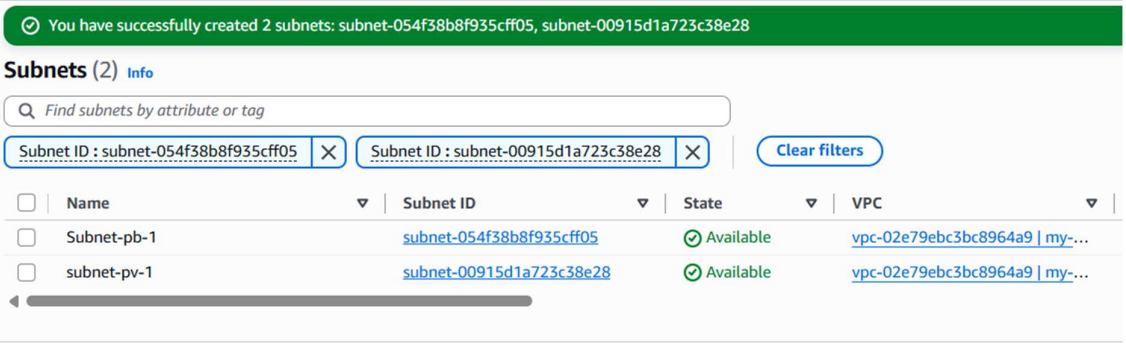
1. Create one VPC, with 1 public subnet and 1 private subnet.

Step 1:- Create a new Vpc in VPC service.

Step 2: -Now selecting the new_VPC create a public and private subnet.





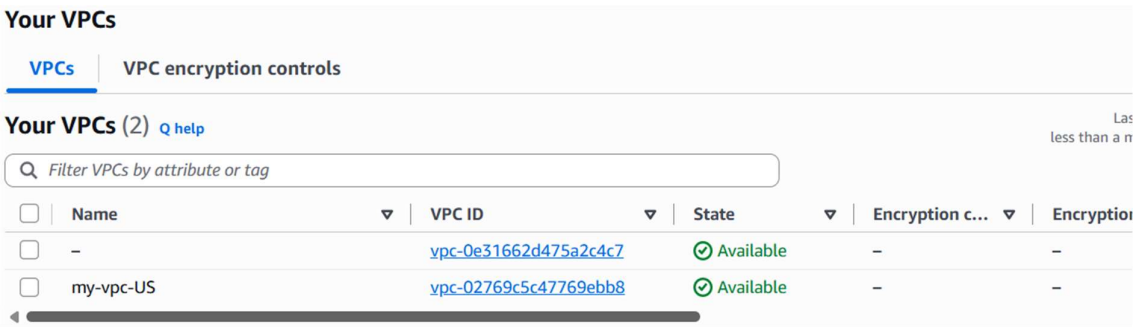


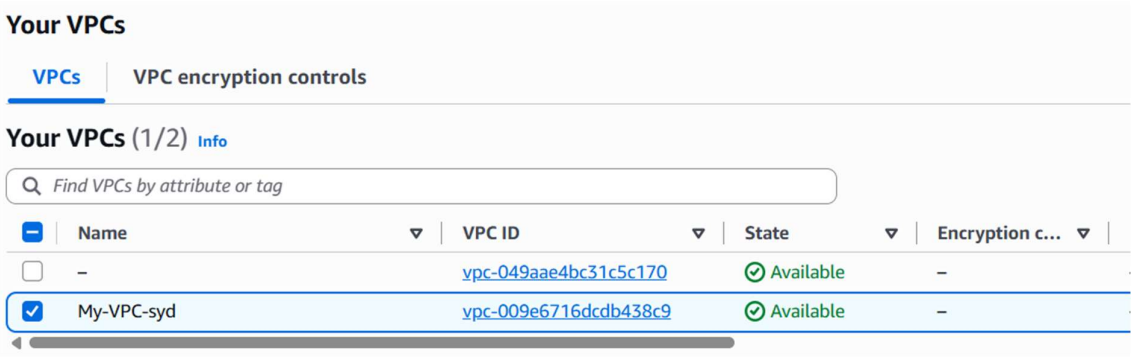
2. Enable VPC peering for cross-region.

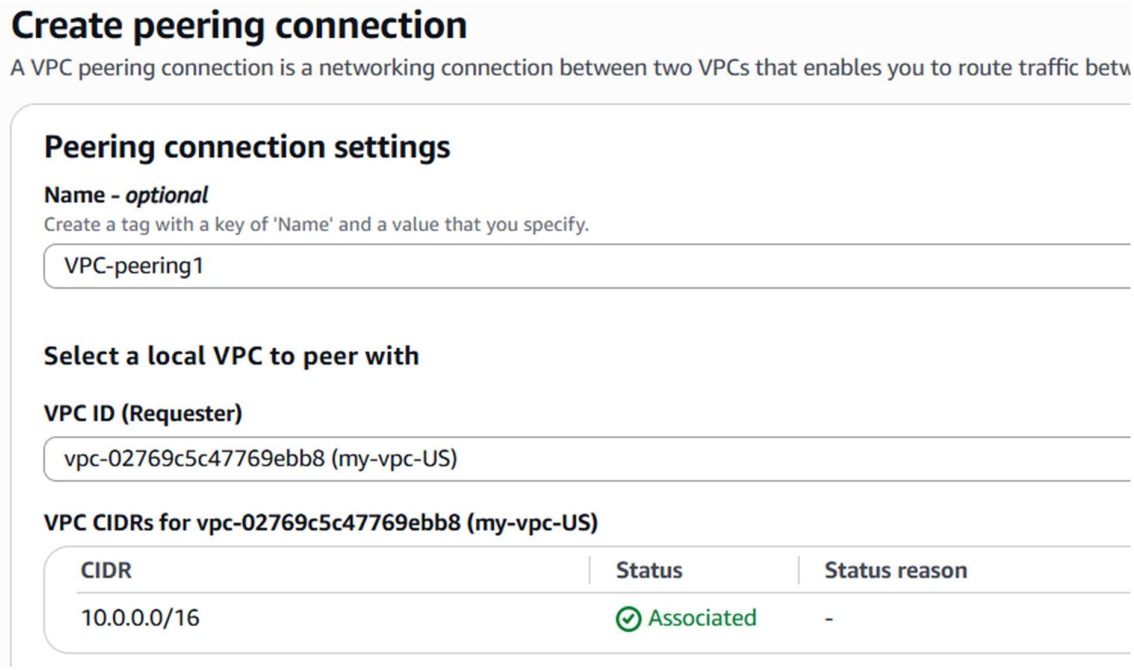
Step 1:- Create VPC, subnet, route table and internet gateway in two regions.

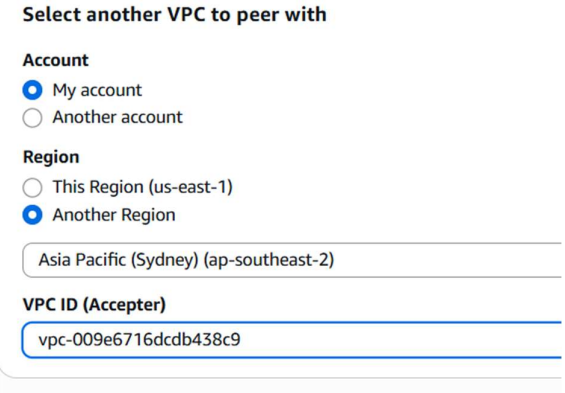
Step 2:- Now create instances, then configure the security groups. Then connect any one of the instance.

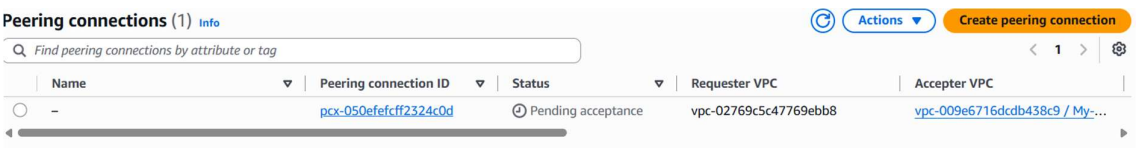
Step 3:- Then check the connectivity with the instance from other region. With Ping command.




Your VPCs
VPCs
VPC encryption controls
Your VPCs (1/2) Info
Find VPCs by attribute or tag
Name
VPC ID
State
Encryption c...
–
vpc-049aae4bc31c5c170
Available
–
My-VPC-syd
vpc-009e6716dcdb438c9
Available
–


Create peering connection
A VPC peering connection is a networking connection between two VPCs that enables you to route traffic betw
Peering connection settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
VPC-peering1
Select a local VPC to peer with
VPC ID (Requester)
vpc-02769c5c47769ebb8 (my-vpc-US)
VPC CIDRs for vpc-02769c5c47769ebb8 (my-vpc-US)
CIDR
Status
Status reason
10.0.0.0/16
Associated
-


Select another VPC to peer with
Account
My account
Another account
Region
This Region (us-east-1)
Another Region
Asia Pacific (Sydney) (ap-southeast-2)
VPC ID (Accepter)
vpc-009e6716dcdb438c9


Peering connections (1) Info
Actions
Create peering connection
Find peering connections by attribute or tag
1
Name
Peering connection ID
Status
Requester VPC
Accepter VPC
–
pcx-050efefcff2324c0d
Pending acceptance
vpc-02769c5c47769ebb8
vpc-009e6716dcdb438c9 / My-...

pcx-050efefcff2324c0d

Actions ▲

- Accept request
- Reject request
- Edit DNS settings
- Manage tags
- Delete peering connection

Pending acceptance
You can accept or reject this peering connection request using the 'Actions' menu. You have until Tuesday, November 10, 2026 at 13:18:14 GMT+5:30 to acc
request, otherwise it expires.

Details Info

Requester owner ID
472567279649

Peering connection ID
pcx-050efefcff2324c0d

Status
Pending Acceptance by 472567279649

Expiration time
Tuesday, November 10, 2026 at 13:18:14 GMT+5:30

Accepter owner ID
472567279649

Requester VPC
vpc-02769c5c47769ebb8

Requester CIDRs
10.0.0.0/16

Requester Region
N. Virginia (us-east-1)

VPC Peering connection AR
arn:aws:ec2:ap-southeast-2:472567279649:vpc-peering-connection/pcx-050efefcff2324c0d

Accepter VPC
vpc-009e6716dcdb438c9 / My-VPC-syd

Accepter CIDRs
–

Accepter Region
Sydney (ap-southeast-2)

DNS | Route tables | Tags

Accept VPC peering connection request Info

Are you sure you want to accept this VPC peering connection request? (pcx-050efefcff2324c0d)

Requester VPC
vpc-02769c5c47769ebb8

Accepter CIDRs
–

Requester owner ID
472567279649(This account)

Accepter VPC
vpc-009e6716dcdb438c9 / My-VPC-syd

Requester Region
N. Virginia (us-east-1)

Accepter owner ID
472567279649(This account)

Requester CIDRs
10.0.0.0/16

Accepter Region
Sydney (ap-southeast-2)

Cancel | Accept request

[Alt+S] | Asia Pacific (Sydney) | Faisal_MOhd

0a5c5964fe28a

Instance summary for i-01b20a5c5964fe28a (Public-syd) Info

Connect | Instance state | Actions

Updated less than a minute ago

Instance ID
i-01b20a5c5964fe28a

IPv6 address
–

Hostname type
IP name: ip-172-21-74-15.ap-southeast-2.compute.internal

Instance type
t3.micro

VPC ID
vpc-009e6716dcdb438c9 (My-VPC-syd)

Subnet ID
subnet-02e77a8d7e4b43e21 (peer-pub)

Instance ARN
arn:aws:ec2:ap-southeast-2:472567279649:instance/i-01b20a5c5964fe28a

Public IPv4 address
54.153.169.177 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-21-74-15.ap-southeast-2.compute.internal

Elastic IP addresses
–

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
| Learn more

Auto Scaling group name
–

Managed
false

Private IPv4 addresses
172.21.74.15

Public DNS
–

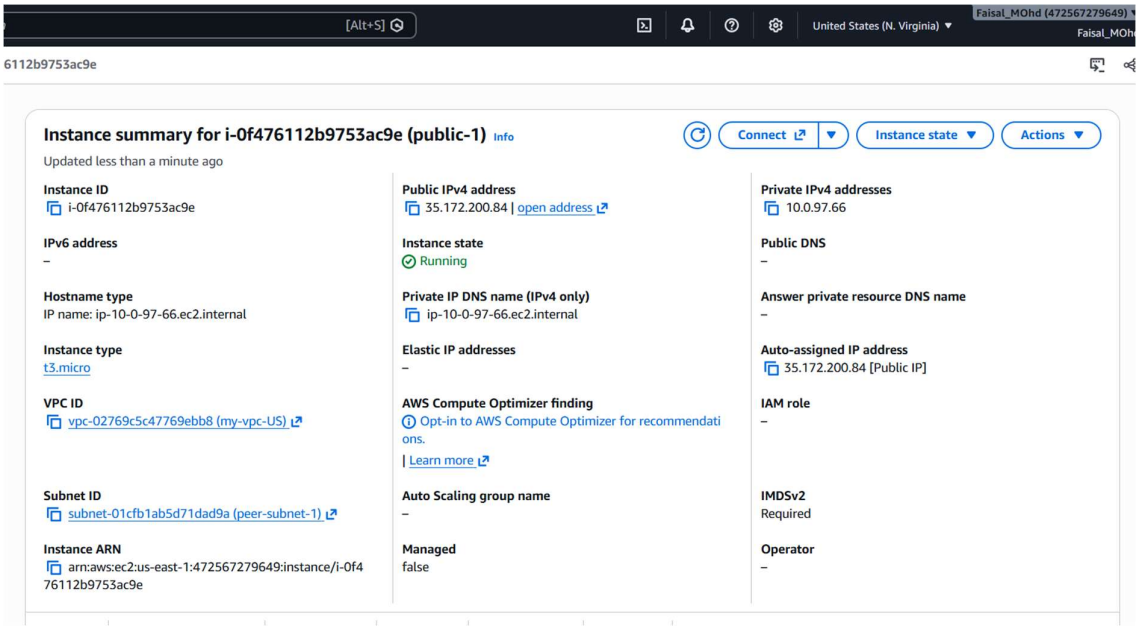
Answer private resource DNS name
–

Auto-assigned IP address
54.153.169.177 [Public IP]

IAM role
–

IMDSv2
Required

Operator
–



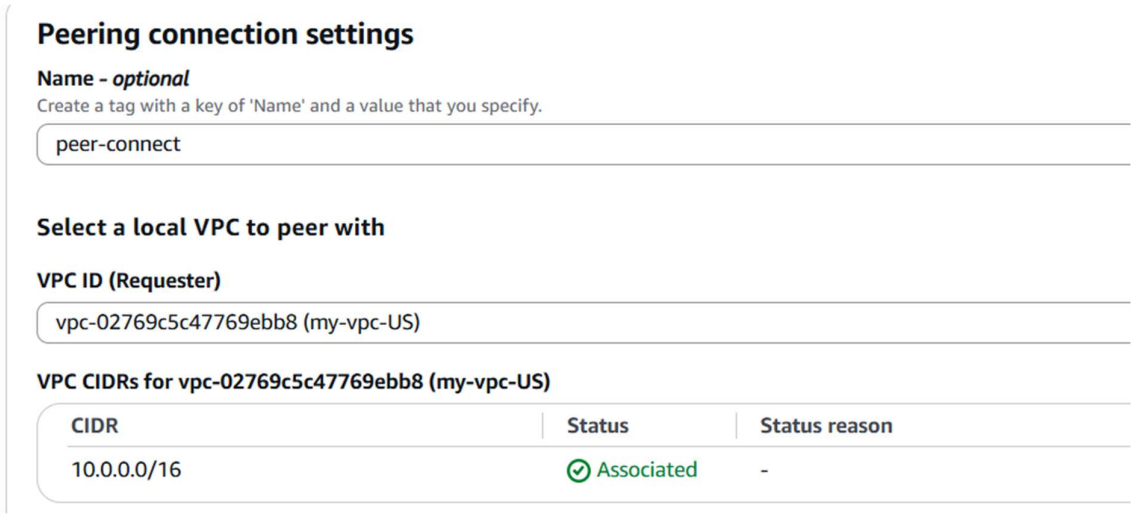

```
Faisal@LAPTOP-4DO8SS4E UCRT64 ~/Downloads
$ ssh -i EC2.pem ec2-user@35.172.200.84
The authenticity of host '35.172.200.84 (35.172.200.84)' can't be established.
ED25519 key fingerprint is: SHA256:m4BzV19mr5ttLIOMXuyaQofqxy7/QDIQ5bL4sfqYw2o
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '35.172.200.84' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-97-66 ~]$ sudo -i
[root@ip-10-0-97-66 ~]# ping 54.153.169.177
PING 54.153.169.177 (54.153.169.177) 56(84) bytes of data.
64 bytes from 54.153.169.177: icmp_seq=1 ttl=117 time=203 ms
64 bytes from 54.153.169.177: icmp_seq=2 ttl=117 time=203 ms
64 bytes from 54.153.169.177: icmp_seq=3 ttl=117 time=203 ms
64 bytes from 54.153.169.177: icmp_seq=4 ttl=117 time=203 ms
64 bytes from 54.153.169.177: icmp_seq=5 ttl=117 time=203 ms
64 bytes from 54.153.169.177: icmp_seq=6 ttl=117 time=203 ms
64 bytes from 54.153.169.177: icmp_seq=7 ttl=117 time=203 ms
64 bytes from 54.153.169.177: icmp_seq=8 ttl=117 time=203 ms
^C
--- 54.153.169.177 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7008ms
rtt min/avg/max/mdev = 203.183/203.206/203.305/0.038 ms
[root@ip-10-0-97-66 ~]#
```

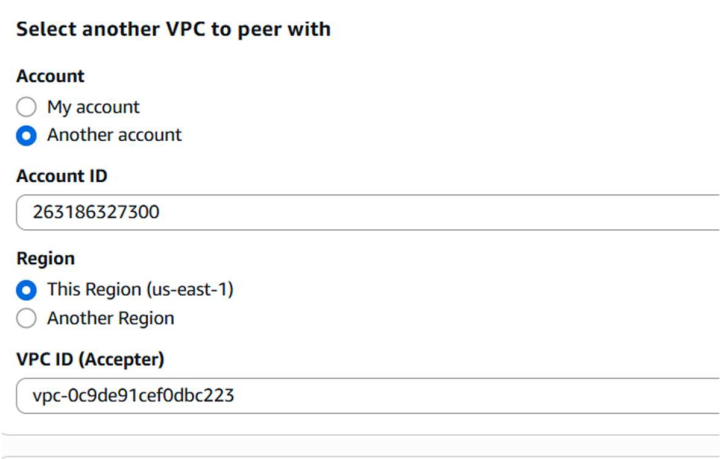
3. Enable VPC peering for cross-account (you can collaborate with your friend to do this task).

Step 1:- Now go to peering connection and create a new peer connection with another user configuration.

Step 2:- Add the other user id, VPC id, and their region.

Step 3:- Then send the request to other user. When they accept the invitation. We can check the connection with Ping command.






```
Faisal@LAPTOP-4DO8SS4E UCRT64 ~/Downloads
$ ssh -i EC2.pem ec2-user@35.172.200.84
The authenticity of host '35.172.200.84 (35.172.200.84)' can't be established.
ED25519 key fingerprint is: SHA256:m4BzV19mr5ttLIOMXuyaQofqxy7/QDIQ5bL4sfqYw2o
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '35.172.200.84' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-97-66 ~]$ sudo -i
```

```
[root@ip-10-0-97-66 ~]# ping 44.215.103.22
PING 44.215.103.22 (44.215.103.22) 56(84) bytes of data.
64 bytes from 44.215.103.22: icmp_seq=1 ttl=126 time=0.651 ms
64 bytes from 44.215.103.22: icmp_seq=2 ttl=126 time=2.53 ms
64 bytes from 44.215.103.22: icmp_seq=3 ttl=126 time=2.10 ms
64 bytes from 44.215.103.22: icmp_seq=4 ttl=126 time=0.749 ms
64 bytes from 44.215.103.22: icmp_seq=5 ttl=126 time=0.629 ms
64 bytes from 44.215.103.22: icmp_seq=6 ttl=126 time=0.663 ms
64 bytes from 44.215.103.22: icmp_seq=7 ttl=126 time=0.646 ms
^C
--- 44.215.103.22 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6173ms
rtt min/avg/max/mdev = 0.629/1.138/2.526/0.753 ms
[root@ip-10-0-97-66 ~]#
```

4. Set up a VPC Transit Gateway.

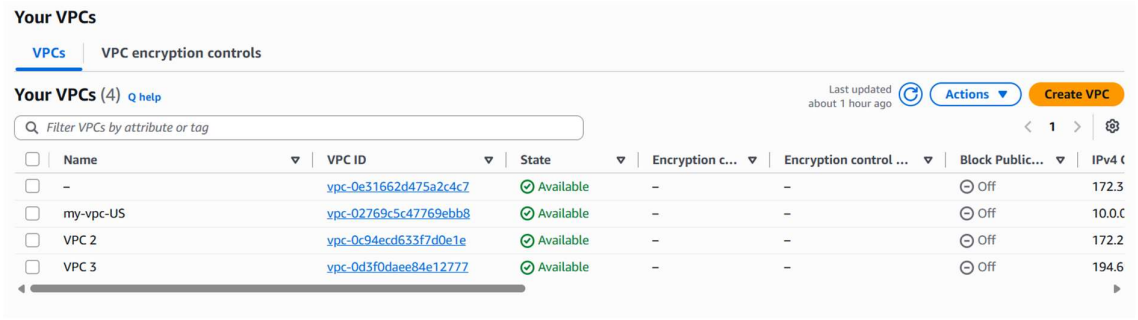
Step 1:- First create 3 VPC, 3 subnets (One public, two private), route tables, one IGW.

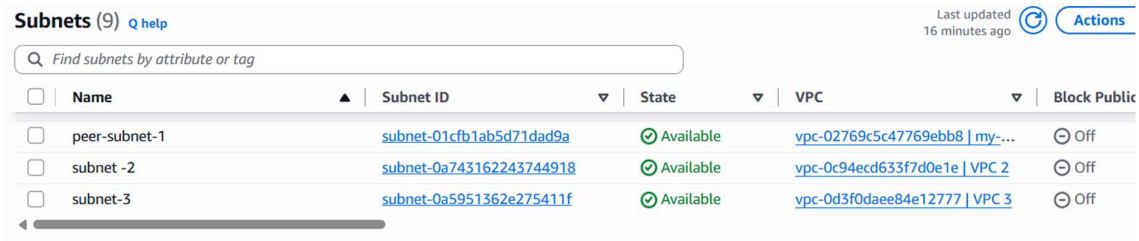
Step 2:- Go to transit gateway and create an Individual Transit gateway, then go to transit gateway attachments and add 3 vpc's.

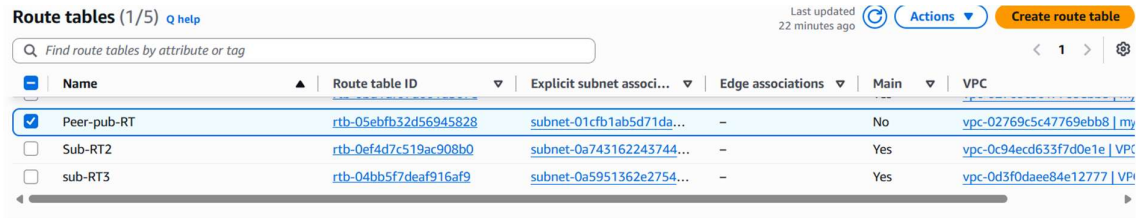
Step 3:- Then add the TGW attachments to route table of all the VPC Routes

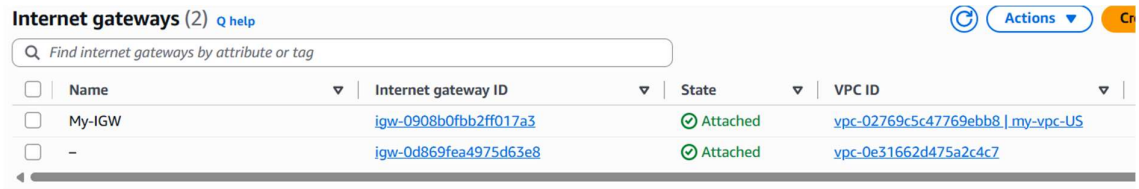
Step 4:- Now create the instances, configure the security groups.

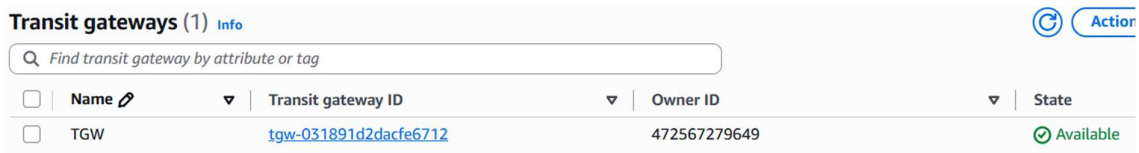
Step 5: - Then to check the connectivity is established or not we can use ping IP address.











Transit gateway attachments (3) Info

Actions

Find transit gateway attachment by attribute or tag

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource ...
vpc-tgw3	tgw-attach-02bf1809491351820	tgw-031891d2dacfe6712	Available	VPC
vpc-tgw2	tgw-attach-0a5233c4726663254	tgw-031891d2dacfe6712	Available	VPC
vpc-tgw1	tgw-attach-0d021d51b562f9df3	tgw-031891d2dacfe6712	Available	VPC

rtb-05ebfb32d56945828 / Peer-pub-RT

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Routes (4)

IP version: Both — Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0908b0fbb2ff017a3	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table
172.21.0.0/16	tgw-031891d2dacfe6712	Active	No	Create Route
194.69.0.0/16	tgw-031891d2dacfe6712	Active	No	Create Route

Instances (3) Info

Last updated 4 minutes ago — Connect — Instance state — Actions — Launch instances

Saved filter sets: Choose filter set

Filter Instance by attribute or tag (case-sensitive)

Instance state = running — Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
private-2	i-064725769ccf93527	Running	t3.micro	3/3 checks passed	us-east-1b	–
private-3	i-054198a0d7119b62b	Running	t3.micro	3/3 checks passed	us-east-1f	–
public-1	i-0f476112b9753ac9e	Running	t3.micro	3/3 checks passed	us-east-1c	–

```
Faisal@LAPTOP-4DO8SS4E UCRT64 ~/Downloads
$ ssh -i EC2.pem ec2-user@35.172.200.84
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Tue Nov  3 08:58:27 2026 from 103.203.172.230
[ec2-user@ip-10-0-97-66 ~]$ ping 194.69.0.152
PING 194.69.0.152 (194.69.0.152) 56(84) bytes of data.
^C
--- 194.69.0.152 ping statistics ---
379 packets transmitted, 0 received, 100% packet loss, time 393148ms

[ec2-user@ip-10-0-97-66 ~]$ ping 194.69.0.152
PING 194.69.0.152 (194.69.0.152) 56(84) bytes of data.
64 bytes from 194.69.0.152: icmp_seq=192 ttl=126 time=2.37 ms
64 bytes from 194.69.0.152: icmp_seq=193 ttl=126 time=0.933 ms
64 bytes from 194.69.0.152: icmp_seq=194 ttl=126 time=0.837 ms
64 bytes from 194.69.0.152: icmp_seq=195 ttl=126 time=0.853 ms
64 bytes from 194.69.0.152: icmp_seq=196 ttl=126 time=0.888 ms
```

```
[ec2-user@ip-10-0-97-66 ~]$ ping 172.21.14.175
PING 172.21.14.175 (172.21.14.175) 56(84) bytes of data.
64 bytes from 172.21.14.175: icmp_seq=1 ttl=126 time=4.12 ms
64 bytes from 172.21.14.175: icmp_seq=2 ttl=126 time=0.894 ms
64 bytes from 172.21.14.175: icmp_seq=3 ttl=126 time=0.802 ms
64 bytes from 172.21.14.175: icmp_seq=4 ttl=126 time=0.834 ms
64 bytes from 172.21.14.175: icmp_seq=5 ttl=126 time=0.858 ms
```

5. Set up a VPC Endpoint.

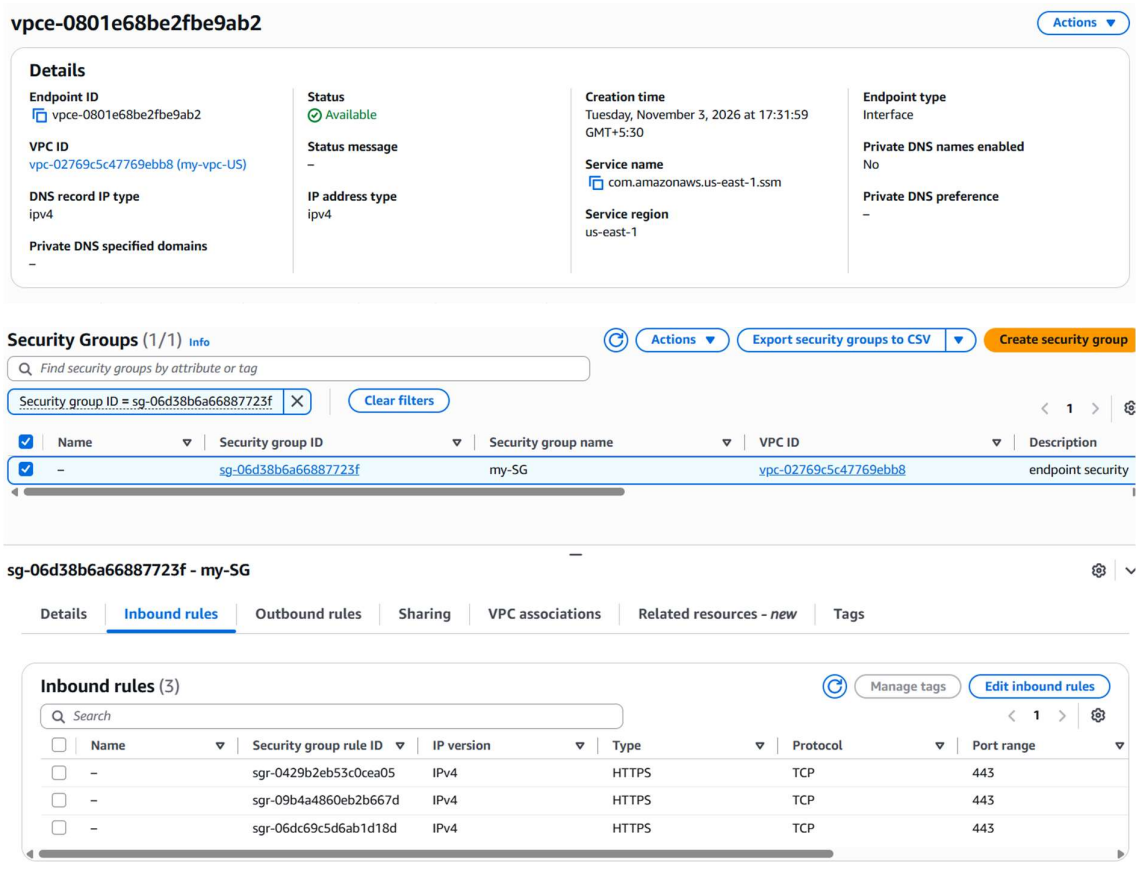
Step 1:- Create a new security group which contains all the VPC Ip address

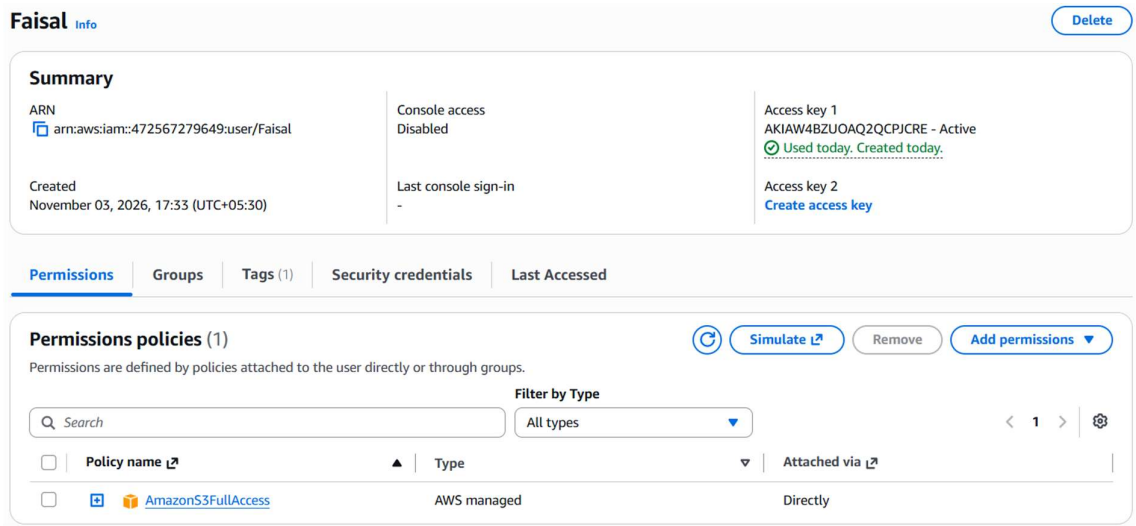
Step 2:- Then create a Endpoint with the main VPC. Now go to IAM user, either create new user or take the access key and secret key from the IAM user.

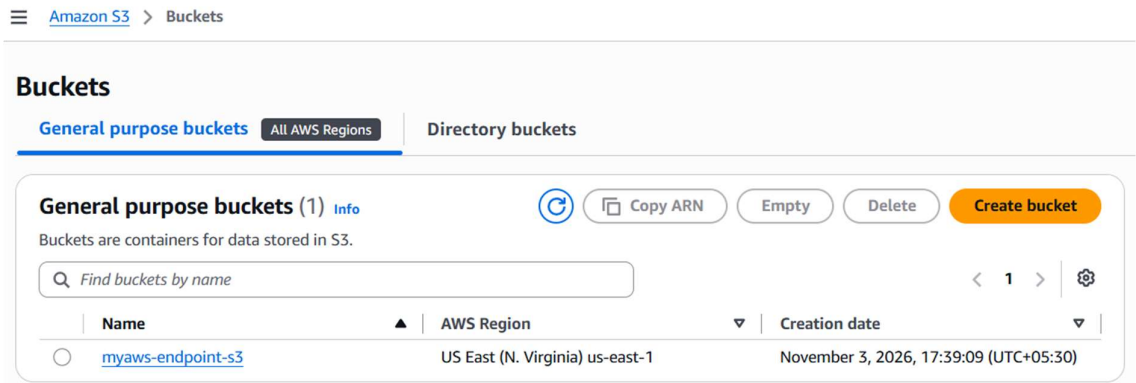
Step 3:- Add amazons3fullaccess policy to the user policies.

Step 4:- Create a s3 bucket in S3 services.

Step 5:- Now configure the credentials in aws cli. Then use aws s3 ls to check the s3 bucket from within the insatance.








```
[ec2-user@ip-10-0-97-66 ~]$ aws --version
aws-cli/2.38.0 Python/3.9.25 Linux/6.18.56-125.164.amzn2023.x86_64 source/x86_64.amzn
[ec2-user@ip-10-0-97-66 ~]$ aws configure

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console sess

AWS Access Key ID [None]: AKIAW4BZUOAQ2QCPJCRE
AWS Secret Access Key [None]: W6nAjQEiTm7oENE6YxOcIfCtplQKdgPKmunV8HOP
Default region name [None]: us-east-1
Default output format [None]: json
[ec2-user@ip-10-0-97-66 ~]$ aws s3 ls

aws: [ERROR]: An error occurred (AccessDenied) when calling the ListBuckets operation
ot authorized to perform: s3:ListAllMyBuckets because no identity-based policy allows
[ec2-user@ip-10-0-97-66 ~]$ aws s3 ls

aws: [ERROR]: An error occurred (AccessDenied) when calling the ListBuckets operation
ot authorized to perform: s3:ListAllMyBuckets because no identity-based policy allows
[ec2-user@ip-10-0-97-66 ~]$ aws s3 ls
2026-11-03 12:09:09 myaws-endpoint-s3
[ec2-user@ip-10-0-97-66 ~]$
```